

Chapter Two: Theory of Demand and Supply

2.1 Theory of Demand

- ❖ The theory of demand is related to the economic activities of consumers-consumption.
- ❖ The purpose of the theory of demand is to determine the various factors that affect demand.

What is demand in Economics?

- ❖ In economics the word “Demand” has a specific meaning, which is different from what we use it in our day to day activities.
- ❖ Demand refers to the amount of commodity which an individual buyer is willing and able to buy at a given price and during a given period of time.

- ❖ Thus, demand is different from **a mere desire**.
- ❖ Human wants are unlimited, and therefore, desires are many. But only a desire that is backed up by the **capacity to pay the price** for the commodity and **the willingness to buy it**, is termed as a **demand**.
- ❖ We may say **demand** refers to an **effective desire/wish**.

Demand = **ability to pay** + **willingness to pay** + **availability of the good**

- ❖ **Law of demand:** This is the **principle of demand**, which states that, **price of a commodity** and its **quantity demanded** are inversely related i.e., as price of a commodity increases (decreases) quantity demanded for that commodity decreases (increases), ceteris paribus.

2.1.1 Demand schedule (table), demand curve and demand function

These are **three ways of representing** the relationship that exists between price and the amount of a commodity purchased.

A) A demand schedule: is the relationship between price and quantity demanded in a **table form**.

Table 2.1: Individual household demand for orange per week

	A	B	C	D	E
Price (Per KG)	5	4	3	2	1
QD (Per Week)	5	7	9	11	13

B) Demand curve: is a graphical representation of the relationship between different quantities of a commodity demanded by an individual at different prices per time period.

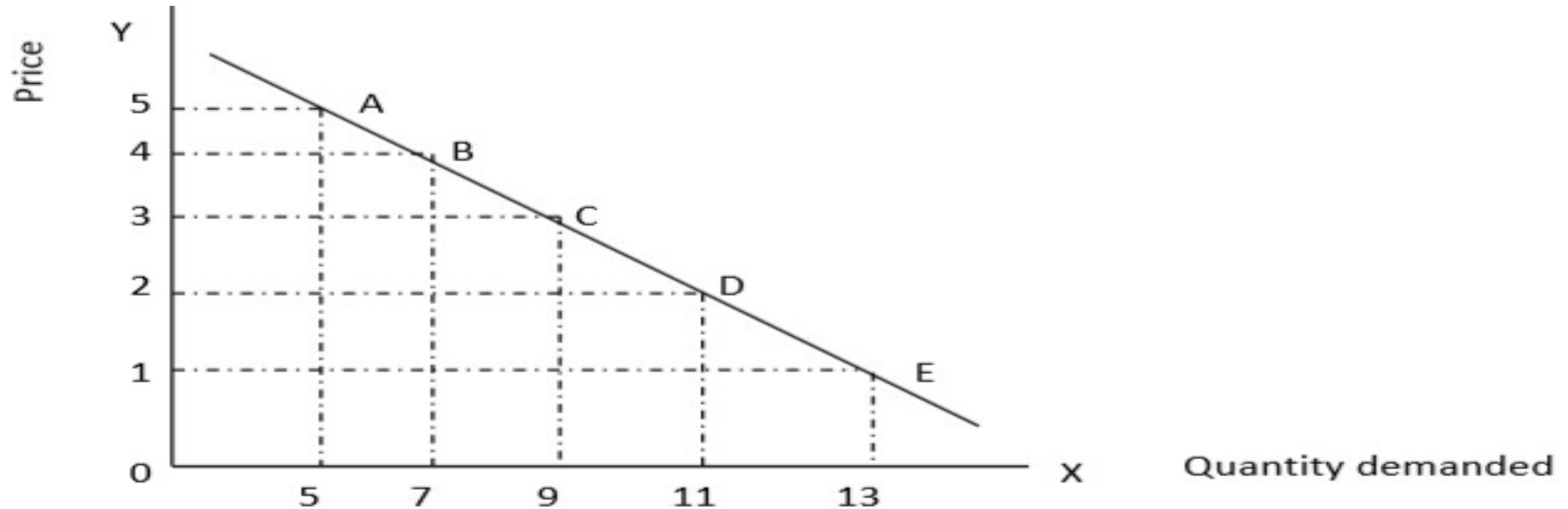


Figure 2.1: Individual demand curve

C) Demand function: is a **mathematical relationship** between price and quantity demanded, all other things remaining the same.

❖ A typical demand function is given by:

$$Q_D = f(p)$$

❖ **Example:** Let the demand function be $Q_D = a + bP$

Where $b = \frac{\Delta Q}{\Delta P}$, which is the slope of the demand curve.

For instance, if we move from point A to point B on figure 2.1 or Table 2.1, then,

$$b = \frac{\Delta Q}{\Delta P} = \frac{7-5}{4-5} = \frac{2}{-1} = \underline{\underline{-2}}$$

Thus, $Q_D = a - 2P$

To find a, let's substitute for Q_D and P at point A or B.

$$7 = a - 2(4); \quad a = 7 + 8 = 15$$

$Q_D = 15 - 2P$: The individual demand function

Market Demand: The **market demand schedule**, **curve** or **function** is derived by **horizontally adding the quantity demanded** for the product by all buyers **at each price**.

Example:

Table 2.2: Individual and market demand for a commodity

Prices	Individual Demands			Market Demand
	Consumer 1	Consumer 2	Consumer 3	
8	0	0	2	0
5	3	5	4	9
3	5	7	6	14
0	7	9	8	20

❖ The following graph depicts market demand curve **at price equal to three**.

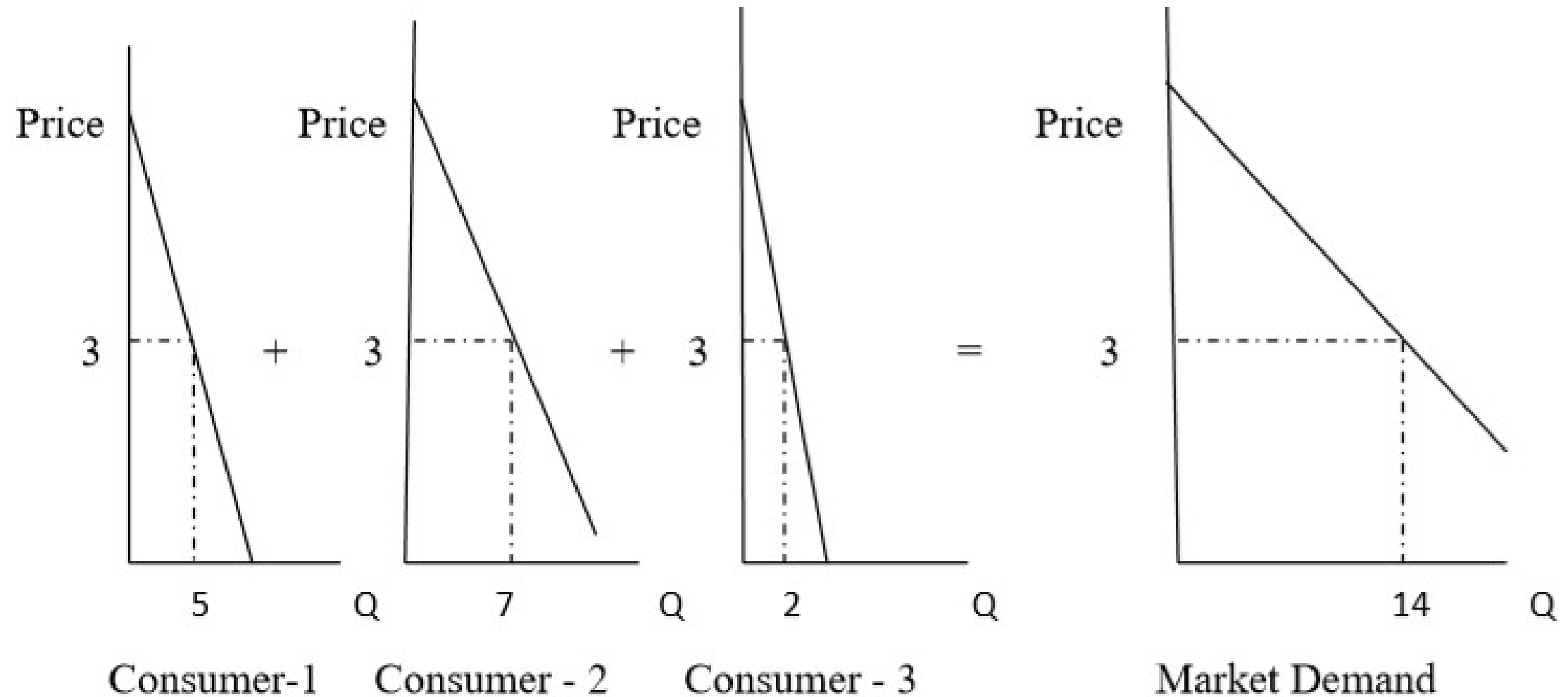


Figure 2.2: Individual and Market demand curve

Market Demand Function:

Example: Suppose the individual demand function of a product is given by:
 $P = 10 - Q/2$ and there are about **100 identical buyers** in the market.

Then the market demand function is given by:

$$P = 10 - \frac{Q}{2}$$

$$\frac{Q}{2} = 10 - P$$

$$Q = 20 - 2P$$

*Market Demand Function = Number of buyers * Individual Demand function*

$$Q_M = (20 - 2P)100$$

$$Q_M = 2000 - 200P$$

Thus, $Q_m = 2000 - 200p$: **Market Demand Function**

2.1.2 Determinants of demand

❖ The demand for a product is influenced by many factors. Some of these factors include:

A) Price of the product itself: The price of a commodity is the most important factor which affects the demand for a commodity.

❖ Other things remaining the same, if price increases, quantity demanded decreases, and if price decreases, quantity demanded increases (**Law of Demand**).

B) Income of the Consumer: Income of the consumer is also an important factor affecting the demand for a commodity. Generally, when income increases, demand also increases, and when income decreases, demand also decreases. This is true in the case of **normal goods**. However, in the case of **inferior goods**, with an increase in income their demand decreases and vice-versa.

On the basis of nature, goods can be classified into two types:

i) Normal Goods (Superior Goods): refer to those goods whose **income effect is positive** – i.e., all other factors remaining the same, as income increases, demand also increases and vice-versa.

For example: Cheese, Butter, Chocolates, Biscuits, etc.

ii) Inferior Goods: Inferior goods refer to those goods whose **income effect is negative** – i.e., all other factors remaining the same, as income increases, demand decreases and vice-versa. In general, inferior goods are **poor quality goods** with relatively lower price and buyers of such goods are expected to shift to better quality goods as their income increases.

For example: Some Chinese shoes, **coarse cloth**, leftover food etc.

C) Prices of Related Goods: Changes in the prices of related goods also affect the demand for a commodity.

❖ Related goods may be of two types:

i) Substitute Goods: are those goods which can be used in place of each other to satisfy a given want. That is why they are also called **competitive goods**.

❖ **For example**, Coffee and tea, Pepsi and Coca-Cola, pens and pencils, butter and oil, etc.

ii) Complementary Goods: are those goods which are used **together/jointly** to satisfy a given want. If two goods are complementary goods, a decline in the price of one would directly change the demand for the other commodity and vice-versa.

❖ **For example**, cars and petrol/fuel, pen and ink, tea and sugar are complements of each other.

D) Tastes and Preferences: If a consumer is accustomed to certain commodities, he will demand that commodity and this leads to increase in the demand for that commodity.

❖ When the taste of a consumer changes in favor of a good, her/his demand will increase and the opposite is true.

E) Consumer expectation of income and price

- ❖ **Higher price expectation** will increase demand while a lower future price expectation will decrease the demand for the good.

F) Number of buyer in the market(Population) and family size

- ❖ Since market demand is the horizontal sum of individual demand, an increase in the number of buyers will increase demand while a decrease in the number of buyers will decrease demand.

G) Climate/Weather: The demand for a commodity is also affected by climate.

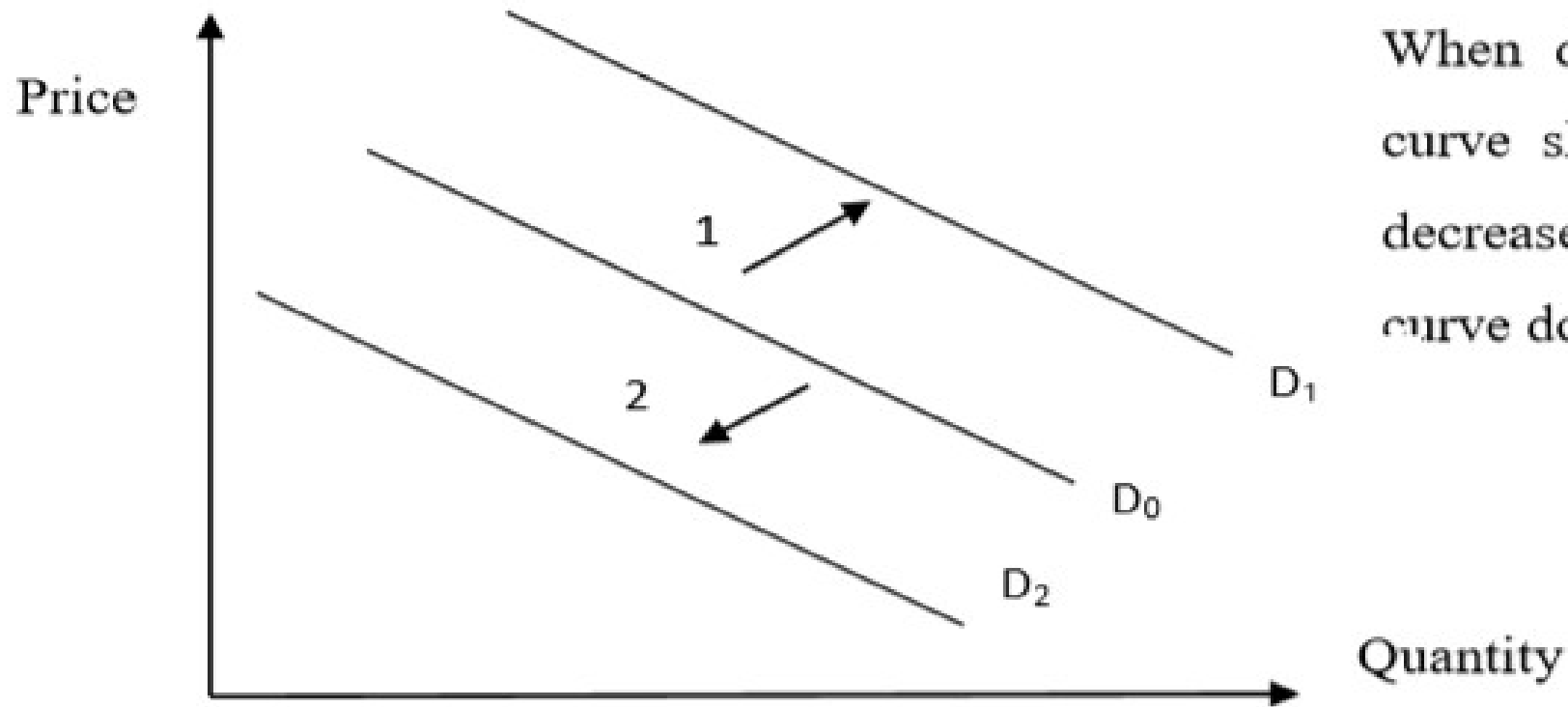
- ❖ **For example**, demand for woolen clothes increases in cold seasons. On the other hand demand for coolers, cotton clothes etc., increases in hot seasons.

Generally, demand mainly depends upon **three factors**, namely. **Price of the commodity**; **Income of the consumer**, and **Price of related goods**.

- ❖ On the basis of the above three factors, demand can be classified into three types: **i) Price Demand**, **ii) Income Demand**, and **iii) Cross Demand**.

Change in Demand

- ❖ a change in any determinant of demand—except for the good's price causes the demand curve to **shift**. We call this **a change in demand**.



When demand increases, demand curve shifts upward (D_1) while a decrease in demand shifts demand curve downwards (D_2).

Figure 2.3: Shift in demand curve

- ❖ When we state **the law of demand**, we kept all the factors to remain constant except the price of the good under consideration.
- ❖ A change in any of the above listed factors except the price of the good will **change the demand**, while a change in the price, other factors remain constant will bring **change in quantity demanded**.
- ❖ A change in demand will shift the demand curve from its original location.
- ❖ For this reason those factors listed above other than price are called **demand shifters**.
- ❖ A change in own price is only a movement along the same demand curve.
- ❖ **Thus, a change in demand** is observed by **a shift in the demand curve**, while **a change in quantity demanded** is expressed by **a movement in the demand curve**.

2.1.3 Elasticity of demand

- ❖ In economics, the concept of elasticity is very crucial and is used to analyze the **quantitative relationship between price and quantity purchased or sold**.
- ❖ **Elasticity** is a measure of responsiveness of a dependent variable to changes in an independent variable.
- ❖ **Elasticity of demand** refers to the degree of responsiveness of quantity demanded of a good to a change in its price, or change in income, or change in prices of related goods.
- ❖ Commonly, there are three kinds of demand elasticity:
 - 1) **price elasticity**,
 - 2) **income elasticity**, and
 - 3) **cross elasticity**.

i) Price Elasticity of Demand

- ❖ **Price elasticity of demand:** refers to the degree of responsiveness of demand to change in price.
- ❖ It is a measure of how much the quantity demanded of a good responds to a change in the price of that good, computed as the percentage change in quantity demanded divided by the percentage change in price.
- ❖ **It indicates how consumers react to changes in price.**
- ❖ The greater the reaction the greater will be the elasticity, and the lesser the reaction, the smaller will be the elasticity.
- ❖ Demand for commodities like **clothes, fruit** etc. changes when there is even a small change in their price, whereas demand for commodities which are basic necessities of life, like **salt, food grains** etc., may not change even if price changes, or it may change, but not in proportion to the change in price.

❖ **Price elasticity demand** can be measured in two ways. These are **point** and **arc elasticity**.

A) Point Price Elasticity of Demand

❖ This is calculated to find elasticity at a given point, and given as:

$$e_d^p = \frac{\text{percentage change in quantity demanded}}{\text{percentage change in price}} = \frac{\% \Delta Q_D}{\% \Delta P}$$

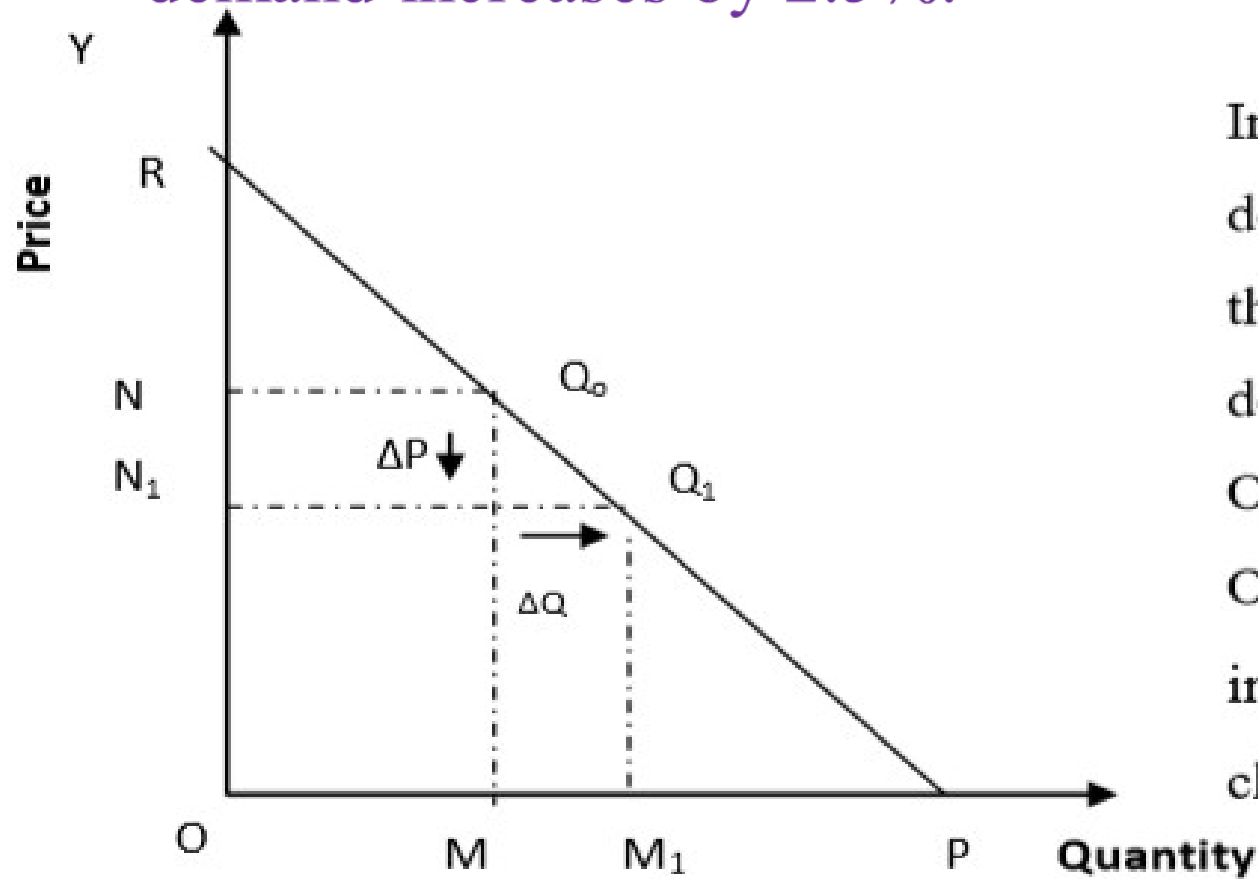
$$e_d^p = \frac{\frac{Q_2 - Q_1}{Q_1} * 100\%}{\frac{P_2 - P_1}{P_1} * 100\%} = \frac{Q_2 - Q_1}{P_2 - P_1} * \frac{P_1}{Q_1} = \frac{\Delta Q}{\Delta P} * \frac{P_1}{Q_1}$$

❖ In this method, we take a straight-line demand curve joining the two axes, and measure the elasticity between two points Q2 and Q1 which are assumed to be intimately close to each other.

❖ **Example:** Suppose the price of the commodity falls from Birr 5 to Birr 4 and quantity demanded increases from 100 units to 150 units. Given this, Compute point price elasticity of demand.

❖ **Solution:** $\frac{Q_2 - Q_1}{P_2 - P_1} * \frac{P_1}{Q_1} = \frac{150 - 100}{4 - 5} * \frac{5}{100} = \frac{250}{100} = -2.5$

❖ This implies that, at price = Birr 5, if price decreases by 1%, quantity demand increases by 2.5%.



In the diagram 'RP' is the straight-line demand curve, which connects both axes. In the beginning at the price ON the quantity demanded is OM. Then the price changes to ON₁ and the new quantity demanded will be OM₁. The symbol 'ΔP' represents the change in price while the symbol 'ΔQ' shows the change in quantity demanded.

Figure 2.4: Point elasticity of demand

NOTE: It should be remembered that **the point elasticity of demand on a straight line is different at every point.**

B) Arc price elasticity of demand

- ❖ The main drawback of the point elasticity method is that it is applicable only when we have information about even the **slight changes in the price** and the quantity demanded of the commodity.
- ❖ But in practice, we do not acquire such information about minute changes. We may possess demand schedules in which there are **big gaps in price as well as the quantity demanded.**
- ❖ In such cases, there is an alternative method known as **arc method** of elasticity measurement.
- ❖ When **elasticity of demand is measured over a finite range** or ‘arc’ of a demand curve, it is called **arc elasticity of demand.**

- ❖ In arc price elasticity of demand, **the midpoints of the old and the new values** of both price and quantity demanded are used.
- ❖ It measures a portion or a segment of the demand curve between the two points.
- ❖ The formula for measuring arc elasticity is given below.

$$\xi_d^a = \frac{\text{Change in quantity demanded}}{\text{Sum of the original and new quantity demanded}} \div \frac{\text{Change in price}}{\text{Sum of the original and new prices}}$$

$$\xi_d^a = \frac{Q_2 - Q_1}{Q_2 + Q_1} \div \frac{P_2 - P_1}{P_2 + P_1} = \frac{\Delta Q}{\Delta P} * \frac{P_2 + P_1}{Q_2 + Q_1}$$

Example: Assuming the previous hypothetical example, compute the arc elasticity of price demand.

Solution: In terms of the above formula,

$$\xi_d^a = \frac{\Delta Q}{\Delta P} * \frac{P_2 + P_1}{Q_2 + Q_1} = \frac{150 - 100}{4 - 5} * \frac{4 + 5}{150 + 100} = \frac{45}{-25} = \frac{-9}{5} = \mathbf{1.8}$$

- ❖ The arc elasticity formula is used if the change in price is **relatively large**.
- ❖ It is a more accurate measure of elasticity than point elasticity method.

❖ From Price elasticity of Demand, we can have the following points

- ✓ Elasticity of demand is **unit free** because it is a ratio of percentage change.
- ✓ Elasticity of demand is **usually a negative number** because of **the law of demand**. If the price elasticity of demand is **positive** the product is **inferior/giffen goods**.
- ✓ If $|\epsilon| > 1$, demand is said to be **elastic** and the product is **luxury** product.
- ✓ If $0 < |\epsilon| < 1$, demand is **inelastic** and the product is **necessity**.
- ✓ If $|\epsilon| = 1$, demand is **unitary elastic**.
- ✓ If $|\epsilon| = 0$, demand is said to be **perfectly inelastic**.
- ✓ If $|\epsilon| = \infty$, demand is said to be **perfectly elastic**.

Determinants of Price Elasticity of Demand

- i) **The availability of substitutes:** the more substitutes available for a product, the more elastic will be the price elasticity of demand.
- ii) **Time:** In the long- run, price elasticity of demand tends to be elastic. Because: More substitute goods could be produced. People tend to adjust their consumption pattern.
- iii) **The proportion of income consumers spend for a product:-**the smaller the proportion of income spent for a good, the less price elastic will be.
- iv) **The importance of the commodity in the consumers' budget :**
 - ✓ **Luxury goods:** tend to be more elastic. **Example:** gold.
 - ✓ **Necessity goods:** tend to be less elastic. **Example:** Salt.
- V) **Number of use of the commodity:**

The higher the number of use of the commodity the higher will be the elasticity.

Example: Electricity.
- vi) **Habits of the consumers:** **Example:** Cigarette Smokers

ii. Income Elasticity of Demand

- ❖ It is a measure of responsiveness of quantity demanded to **change in income**.

$$\epsilon_d^I = \frac{\frac{Q_2 - Q_1}{Q_1} * 100\%}{\frac{M_2 - M_1}{M_1} * 100\%} = \frac{Q_2 - Q_1}{M_2 - M_1} * \frac{M_1}{Q_1} = \frac{\Delta Q}{\Delta M} * \frac{M_1}{Q_1}$$

- ❖ Accordingly,

- ✓ If $\epsilon_d^I > 1$, the good is **luxury good**.
- ✓ If $1 < \epsilon_d^I < \infty$, (and positive), the good is **necessity good**
- ✓ If $\epsilon_d^I < 0$, (negative), the good is **inferior good**.

Example: Suppose a consumer has money income of Birr 1000 and he purchases 4 kg of wheat. If his money income goes up to Birr 1200, he is now prepared to buy 5 kg of wheat. Compute the point income elasticity of demand.

Solution:

$\epsilon_d^I = \frac{Q_2 - Q_1}{M_2 - M_1} * \frac{M_1}{Q_1} = \frac{5 - 4}{1200 - 1000} * \frac{1000}{4} = \frac{1000}{400} = 1.25$, implies for a 1 percent increase in income there is a 1.25 percent increase in the demand of the commodity and the commodity is normal(luxury).

iii) Cross Elasticity of Demand

- ❖ Measures how much the demand for a product is affected by a change in the price of another good(related good).
- ❖ The formula used to compute cross elasticity is:

$$\epsilon_d^{xy} = \frac{\frac{Q_{x2} - Q_{x1}}{Q_{x1}} * 100\%}{\frac{P_{y2} - P_{y1}}{P_{y1}} * 100\%} = \frac{Q_{x2} - Q_{x1}}{P_{y2} - P_{y1}} * \frac{P_{y1}}{Q_{x1}} = \frac{\Delta Q_X}{\Delta P_Y} * \frac{P_{Y1}}{Q_{X1}}$$

According to the values of ξ_d^{xy} ,

- i) If ξ_d^{xy} is **positive**, the goods are **substitute goods**.
- ii) If ξ_d^{xy} is **negative**, the goods are **complementary goods**.
- iii) iii) If ξ_d^{xy} is **zero**, the goods are **unrelated goods**.

Example: Suppose that when the price of a good Y increases from 10 birr to 15 birr, then the quantity demanded of a good X has decreased from 1500 units to 1000 units. Compute the cross price elasticity of demand.

Solution:

$\xi_d^{xy} = \frac{Q_{x2} - Q_{x1}}{P_{y2} - P_{y1}} * \frac{P_{y1}}{Q_{X1}} = \frac{1000 - 1500}{15 - 10} * \frac{10}{1500} = \frac{-500}{5} * \frac{1}{150} = -0.667$, implying for a percent increase in the price of a good Y, there is 0.667 percent decrease in the quantity demanded of price good X. The two good are complementary.

2.2 THEORY OF SUPPLY

- ❖ Supply indicates various quantities of a product that sellers (producers) are willing and able to provide at different prices in a given period of time, other things remaining unchanged.
- ❖ The law of supply: states that, **ceteris paribus**, as price of a product increase, quantity supplied of the product increases, and as price decreases, quantity supplied decreases.

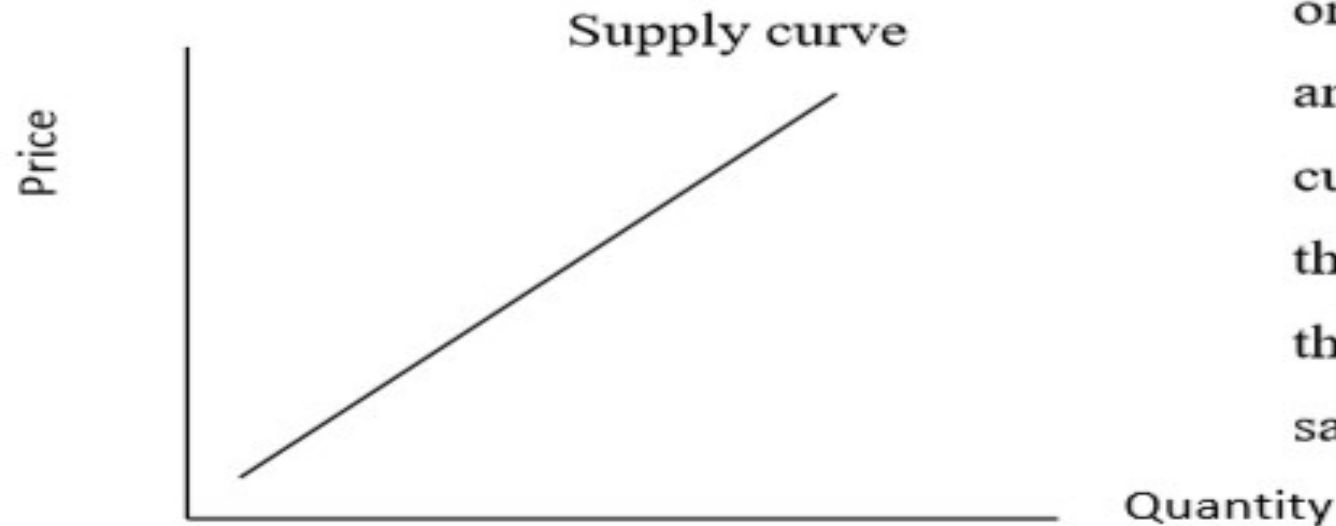
2.2.1 Supply schedule, supply curve and supply function

- ❖ A supply schedule is a tabular statement that states the different quantities of a commodity offered for sale at different prices.

Table 2.3: an individual seller's supply schedule for **butter**

Price (birr per KG)	30	25	20	15	10
QS(KG/Week)	100	90	80	70	60

- ❖ **A supply curve:** conveys the same information as a supply schedule. But it shows the information graphically rather than in a tabular form.



In this diagram the quantities of oranges are measured along X axis and prices along Y axis. The supply curve slopes upward as we go from the left to the right. This means, as the price rises, more is offered for sale and vice-versa.

Figure 2.5 supply curve

- ❖ **Supply Function: Mathematical representation.** The supply function of a commodity can be briefly expressed in the following functional relationship:

$$S = f(P),$$

Where S is quantity supplied and P is price of the commodity

Market supply: It is derived by **horizontally adding the quantity supplied** of the product by all sellers at each price.

Table 2.4: Derivation of the market supply of good X

Price per unit	Quantity supplied by seller 1	Quantity supplied by seller 2	Quantity supplied by seller 3	Market supply per week
5	11	15	8	34
4	10.5	13	7	30.5
3	8	11.5	5.5	25
2	6	8.5	4	18.5
1	4	6	2	12

2.2.2 Determinants of supply

❖ Apart from the change in **price** which causes a change in quantity demanded, the supply of a particular product is determined by:

i) Input Price:

❖ An **increase in the price of inputs** such as labour, raw materials, capital, etc. causes **a decrease in the supply** of the product which is represented by a leftward shift of the supply curve.

ii) State of Technology

❖ Technological advancement enables a firm to produce and supply more in the market. This shifts the supply curve outward.

iii) Price of Related Goods: An increase in the price of other, related goods induces the firms to produce more of those other goods, leading to a reduction in the supply of the goods whose price has remained unchanged.

iv) Objectives of the Firm: Beside/apart from to the primary profit maximization objective, firms could have such as objectives of **maximum sales, maximum employment, more production**, etc. In this case, the supply will be increasing.

V) Weather condition

- ❖ A change in weather condition will have an impact on the supply of a number of products, especially **agricultural products**.

Vi) Sellers' expectation of price of the product:

vii) Number of sellers in the market

vii) Taxes & Subsidies (Fiscal Policy)

viii) Other factors: Market access (infrastructural development), political stability. etc.

2.2.3 Elasticity of supply

- ❖ It is the **degree of responsiveness of the supply to change in price**. It may be defined as the percentage change in quantity supplied divided by the percentage change in price.
- ❖ As the case with price elasticity of demand, we can measure the price elasticity of supply using point and arc elasticity methods.
- ❖ However, a simple and most commonly used method is **point method**.

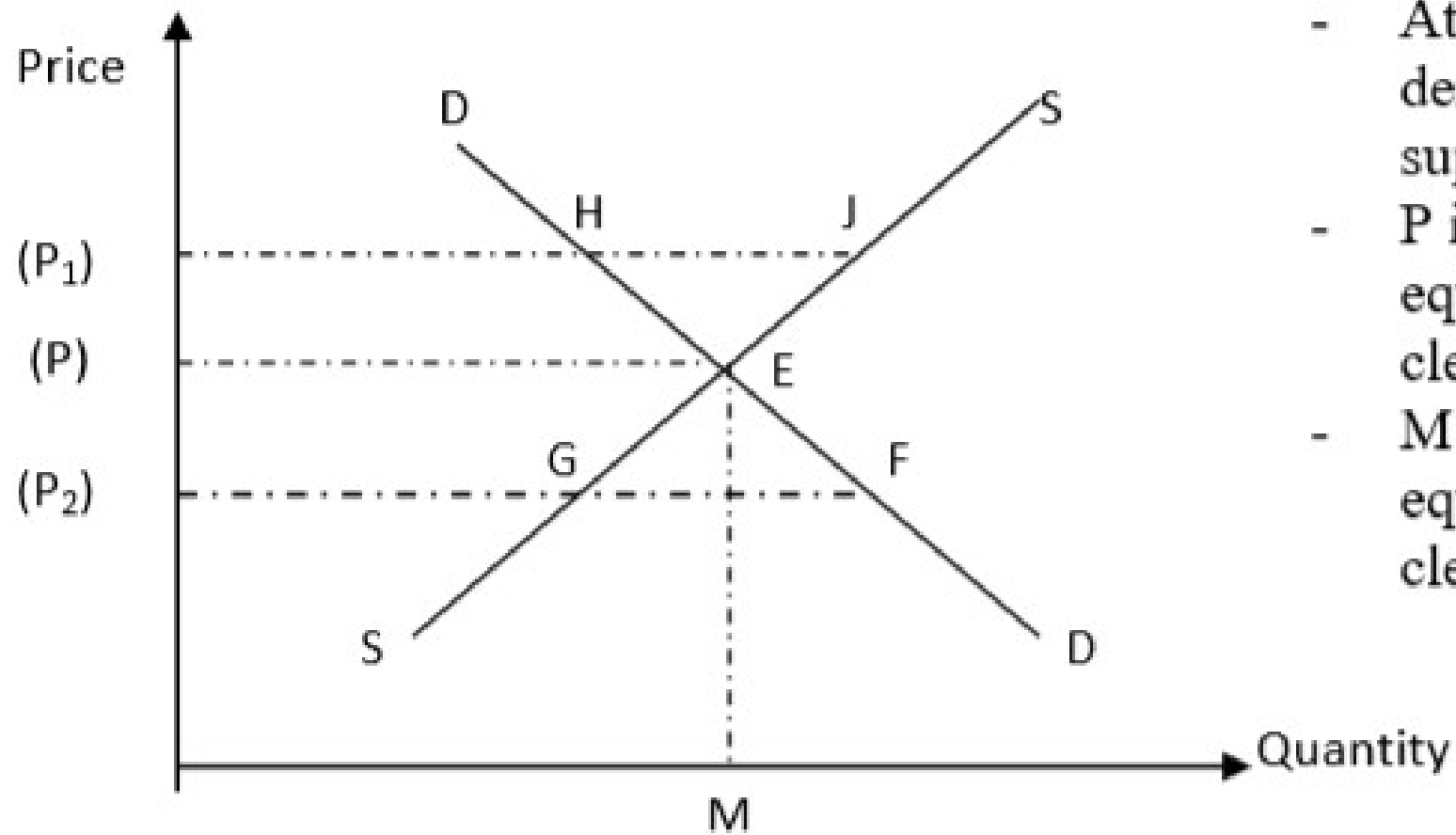
$$\epsilon_s^p = \frac{\text{percentage change in quantity supplied}}{\text{percentage change in price}} = \frac{\% \Delta Q_s}{\% \Delta P}$$

$$\epsilon_s^p = \frac{\frac{Q_2 - Q_1}{Q_1} * 100\%}{\frac{P_2 - P_1}{P_1} * 100\%} = \frac{Q_2 - Q_1}{P_2 - P_1} * \frac{P_1}{Q_1} = \frac{\Delta Q}{\Delta P} * \frac{P_1}{Q_1}$$

- ❖ Given the value of ϵ_s^p , like elasticity of demand, price elasticity of supply can be **elastic**, **inelastic**, **unitary elastic**, **perfectly elastic** or **perfectly inelastic**.

2.3 Market equilibrium

Market equilibrium occurs when **market demand = market supply**.



- At point 'E' market demand equals market supply (equilibrium point)
- P is the market equilibrium (market clearing) price.
- M is the market equilibrium (market clearing) quantity.

Figure 2.7: market equilibrium

Example: Given market demand: $Q_d = 100 - 2P$, and market supply:
 $P = (Q_s / 2) + 10$

- a) Calculate the market equilibrium price and quantity
- b) Determine, whether there is surplus or shortage at $P = 25$ and $P = 35$

Solution:

a) At equilibrium, $Q_d = Q_s$

$$100 - 2P = 2P - 20$$

$$4P = 120$$

$$P^* = 30 \text{ and } Q^* = 40$$

b) $Q_d(\text{at } P = 25) = 100 - 2(25) = 50$ and

$$Q_s(\text{at } P = 25) = 2(25) - 20 = 30$$

Therefore, there is a **shortage** of: $50 - 30 = 20$ units

$$Q_d(\text{at } P = 35) = 100 - 2(35) = 30 \text{ and}$$

$$Q_s(\text{at } p = 35) = 2(35) - 20 = 50$$

Therefore, there is a **surplus** of: $30 - 50 = -20$ units.

Effects of shift in demand and supply on equilibrium

What will happen to the equilibrium price and quantity ?

i) When demand changes and supply remains constant

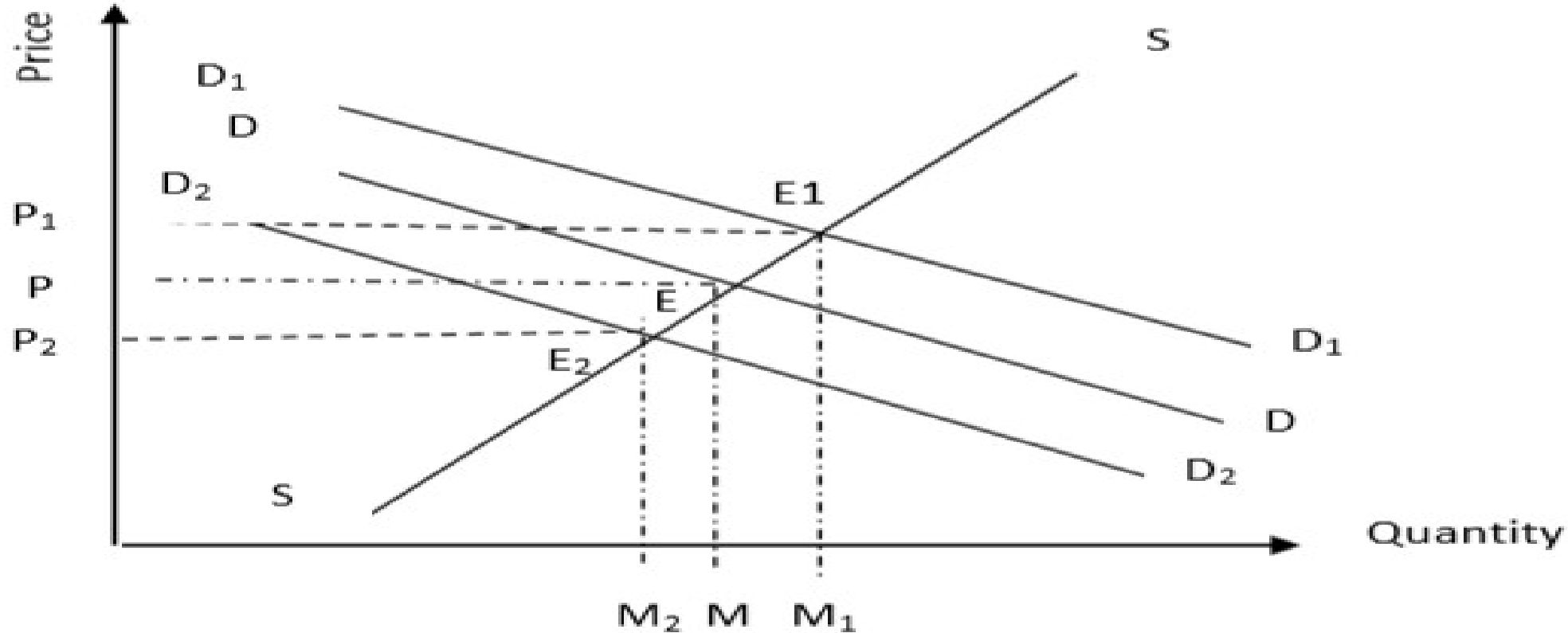


Figure 2.8: The effect of change in demand on market equilibrium

❖ Thus, supply being given, a decrease in demand reduces both the equilibrium **P** and **Q** and vice versa.

ii. When supply changes and demand remains constant

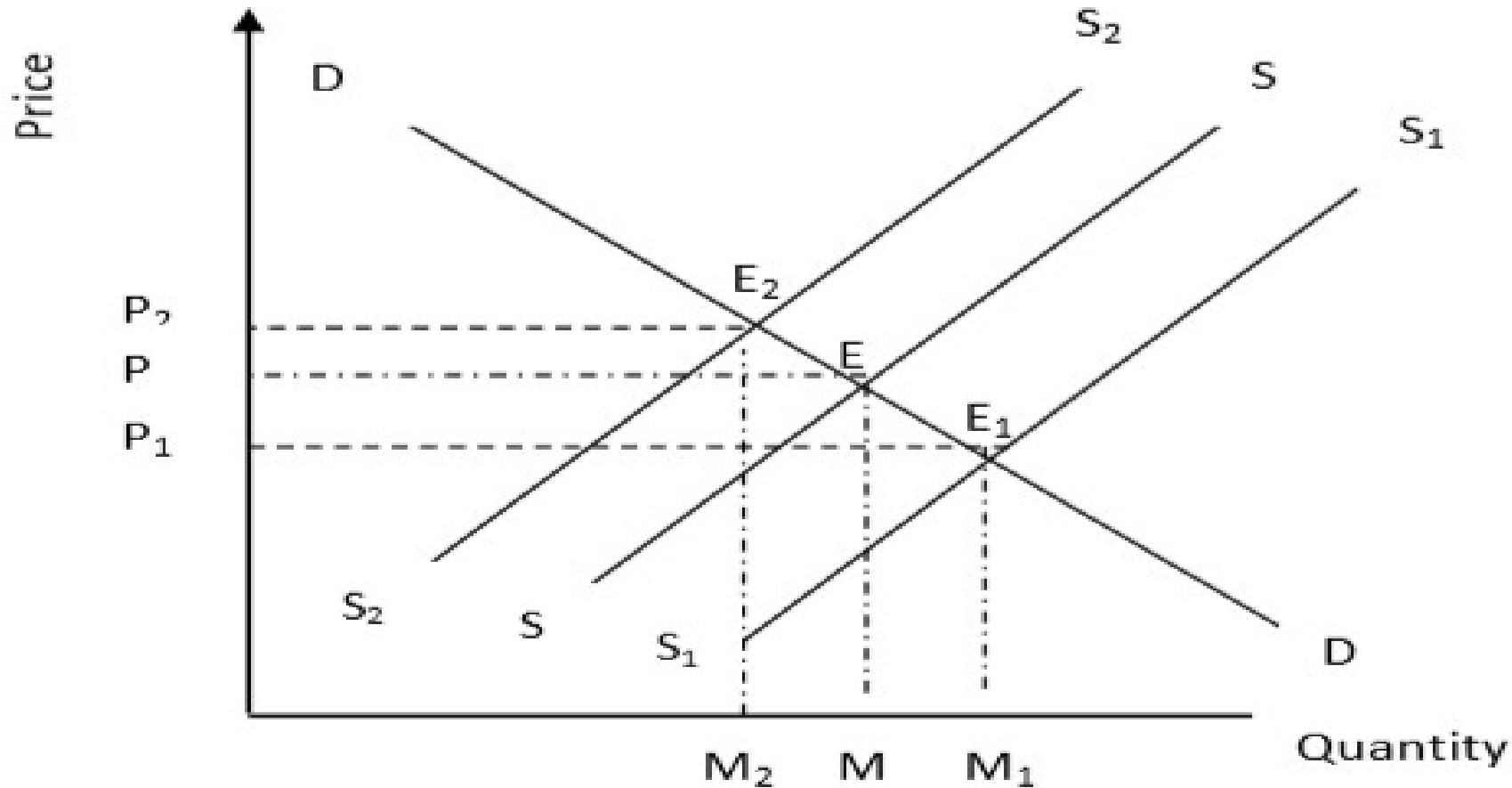


Figure 2.9: The effect of change in supply on market equilibrium

❖ **Thus**, give the demand, an increase in supply **reduces the equilibrium P** and **increases the equilibrium Q**, and vice versa.

III) Effects of combined changes in demand and supply

- ❖ When both demand and supply **increase**, the quantity of the product **will increase definitely**. But it is **not certain** whether the **price** will rise or fall.
- ❖ **Three Scenarios:**
 - 1) If an increase in **demand** is more than an increase in supply, then the **price goes up**.
 - 2) If an increase in **supply** is more than an increase in demand, the **price falls**.
 - 3) If the increase in **demand and supply** is same, then the **price remains the same**.
- ❖ Besides, when demand and supply **decline**, the **quantity decreases**.
- ❖ But it will depend upon **the relative fall in demand and supply**.
change in price

❖ **In this case too, there will be three Scenarios:**

- 1) When the **fall in demand is more than the fall in supply**, the **price will decrease**.
- 2) When the **fall in supply is more than the fall in demand**, the **price will rise**.
- 3) If both demand and supply decline in **the same ratio**, there is **no change in the equilibrium price**, but the quantity decreases.

Therefore, when both supply and demand change, the effect on the equilibrium price depends on the proportion of change (relative change) in demand and change in supply.

Quiz(10%)

I) The market demand for a product is given as: $p = 20 - 1/4Q_d$ and the market supply for the product is given as: $p = \frac{1}{5}Q_s + 2$.

- A) Compute the market clearing price and market clearing quantity. (4%)
- B) What happens to the equilibrium levels of price and quantity in (A),
- i) if both market demand and market supply decline in the same ratio or proportion. (2%)
 - ii) If the decline in market supply is more than the decline in market demand. (2%)
 - iii) If the decline in market demand is more than the decline in market supply. (2%)

NB: Properly demonstrate your Answers for questions in (B) using Graphs.
(Otherwise, will not be evaluated)